

NANYANG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

MATHEMATICS

Higher 1

Paper 1

8863/1

8 September 2008

3 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and name of subject tutors in the space provided.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

Name: _____

Class: _____

Subject Tutor(s): _____

Question No.	Marks	Question No.	Marks	Question No.	Marks
1		5		9	
2		6		10	
3		7		11	
4		8		12	
				Total	100

This question paper consists of **6** printed pages and **2** blank pages.

[Turn over

Section A : Pure Mathematics [40 marks]

- 1 (a) Given that $e^{3x+2} = b^x$, where $b > 0$ and $b \neq e^3$. Find x in terms of b . [2]
 (b) Solve the equation $\lg y + \lg(2y - 3) = 1 + \lg 2$. [3]

- 2 (a) Find, using an algebraic method, the solution set to the inequality $x^2 - x + 1 > 0$. [3]
 (b) Find the range of values of p for which the quadratic expression $px^2 + (p + 3)x + p$ is positive for all real values of x . [5]

- 3 The functions f and g are defined by

$$f : x \mapsto \frac{1}{2x+1}, \quad x > -\frac{1}{2}$$

$$g : x \mapsto e^{x^2}, \quad x \in \mathbb{R}$$

- (i) Define f^{-1} in similar form. [3]
 (ii) Describe a single transformation that will transform the graph of $y = f(x)$ to the graph of $y = f^{-1}(x)$. [1]
 (iii) State the range of g and hence show that fg exists. [2]
 (iv) Define the composite function fg in a similar form. [2]
- 4 (a) The gradient at any point (x, y) of a curve is given by $\frac{dy}{dx} = -\frac{25}{(x+1)^2}$. Given that the curve passes through the points $(0, 25)$ and $(4, k)$, find the value of k . [4]
 (b) Without the use of a calculator, solve the equation $e^{2x} - e^x - 2 = 0$. [2]
 Hence, find the exact area of the region bounded by the curves with equations $y = e^x$, $y = 2e^{-x} + 1$ and the y -axis. [3]

[Turn over]

- 5 (a) Differentiate $\ln\left(\frac{81}{x^2}\right)$ with respect to x . [2]
- (b) The equation of a curve is $y = ae^{2x}$, where a is a constant. Find the equation of the tangent to the curve at $x = 1$. [4]
- (c) A petrol tanker is damaged in a road accident, and petrol leaks onto a flat section of a motorway. The leaking petrol begins to spread in a circle of thickness 2 mm. Petrol is leaking from the tanker at a rate of $0.0084 \text{ m}^3\text{s}^{-1}$. Find the rate at which the radius of the circle of petrol is increasing at the instant when the radius of the circle is 3m, giving your answer in m s^{-1} to 2 decimal places. [4]

Section B : Statistics [60 marks]

- 6 A light bulb manufacturer wishes to test the life-span of the bulbs his company produces by taking a sample.
- (i) Give a reason why it is unsuitable to use simple random sampling. [1]
- (ii) Assume that the manufacturer has 200 light bulbs, of which 40 are “energy saving”, 60 are “long lasting”, and the remaining are “super bright.” Describe how he could select a stratified sample of 50 bulbs for testing. [3]
- 7 (a) The masses of 250 randomly selected apples in an orchard are measured and the results are summarized by $\sum (x - 50) = 8090$ and $\sum (x - 50)^2 = 468308$, where x denotes the mass of an apple in grams. Find the unbiased estimates of the population mean and variance of the apples in the orchard, giving your answers correct to 2 decimal places. [2]
- (b) The random variable X has mean 30 and variance 15. Use the Central Limit Theorem to estimate the probability that the mean of a random sample of 50 observations of X exceeds 29.5. [2]

[Turn over]

- 8** A school health specialist is requested to conduct a test to determine if the mean height of 10-year old students in a particular school is less than the national average height of 124 cm for students of the same age. The heights of 80 randomly chosen 10-year old students in the school are taken and the results, x cm, are summarized by $\sum x = 9740$.

- (i)** Carry out the test at the 5% level of significance, assuming that the population variance of heights of 10-year old students is known to be 120 cm^2 . Give your conclusion in the context of the question. **[4]**
- (ii)** Explain briefly if your conclusion would be different if a two-tail test at the 5% level of significance were to be used instead. **[1]**
- (iii)** State, with a reason, whether it is necessary for the heights to have a normal distribution for the test to be valid. **[1]**

- 9** A study was conducted to determine if there is any correlation between the development of psychomotor skills and speech of a 2-year old toddler. Ten 2-year old toddlers were put through a psychomotor skills test and a speech test each. The scores (maximum 10 for each test) were tabulated.

Toddler	A	B	C	D	E	F	G	H	I	J
Psychomotor skills score, x	8	1	5	10	3	9	2	4	3	1
Speech test score, y	5	5	7	3	10	4	1	8	9	10

- (i)** Give a sketch of the scatter diagram for the data. **[2]**
- (ii)** Calculate the linear product moment correlation coefficient. **[1]**
- (iii)** Calculate the regression lines of y on x and of x on y . **[2]**
- (iv)** Use the appropriate regression line to estimate the following:
 - (a)** the score of a toddler's speech test if the psychomotor test score is 6.5,
 - (b)** the psychomotor test score of a child who scores 0.5 for speech test. **[2]**
- (v)** Comment on the reliability of the estimates in part **(iv)**. **[2]**

- 10** An experiment is performed with a die and two packs of cards, *A* and *B*. The die is thrown, and if it shows 1, 2, 4 or 5, a card is drawn at random from pack *A*, which contains the usual 52 cards. If the score on the die is 3 or 6, a card is drawn from pack *B*, which contains only 26 cards as all the spades and clubs have been removed.

X is the event that a card is drawn from pack *A*.

Y is the event that the card drawn is a diamond picture card.

Draw a tree diagram to illustrate the above events. [2]

Find the following probabilities:

- (a) $P(X)$ [1]
- (b) $P(X \cap Y)$ [1]
- (c) $P(Y)$ [2]
- (d) $P(Y|X)$ [2]
- (e) $P(X|Y)$ [2]

- 11** A bus operator running one route kept a long-term record of journey times and found that, a bus was late (i.e. took longer than timetabled) at the end of the journey once in three times.

(i) On a morning when 8 journeys are made,

- (a) what is the probability that a bus will be late at most 2 times? [2]
- (b) what is the probability that a bus is always on time? [1]

(ii) On a day when 50 journeys are made, using an appropriate approximation, what is the probability that the bus will be late not more than 20 times? [5]

(iii) In a month consisting of 30 journeys, the bus service will be termed as “bad” when the bus is late for more than 10 times. What is the expected number of times that the bus service will be termed as “bad” in a year? [4]

[Turn over]

- 12** The amount of time for junior college girls to run 2.4 km is normally distributed with mean running time at 15 minutes and 30 seconds and standard deviation 2 minutes and 30 seconds. The amount of time for junior college boys to run 2.4 km is also normally distributed with mean running time of 10 minutes and standard deviation 1 minute and 30 seconds.

Calculate the probability that

- (i) a randomly chosen junior college girl takes less than 14 minutes and 20 seconds to complete the 2.4 km run, [2]
- (ii) if the timing of 2 randomly chosen junior college boys are taken, one junior college boy takes more than 12 minutes to complete the 2.4 km run and the other takes less than 9 minutes, [3]
- (iii) a junior college girl can outrun a junior college boy for the 2.4 km run, [3]
- (iv) the mean running time for a sample of 8 girls is less than 15 minutes. [3]

It is known that the girls were given intensive running training during the June holidays. Find the new mean running time for the girls, if the probability that the running time of a randomly chosen junior college girl is now less than 13 minutes is 0.55. (Assume that the standard deviation for the running time remains unchanged.) [4]

End of Paper